

ChainMark

A Cryptographic Breadcrumb Architecture for CSAM Distribution Network Interdiction

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Abstract

Existing tools for combating child sexual abuse material (CSAM) — primarily perceptual hash databases such as PhotoDNA — are effective at identifying known content but are architecturally passive: they detect individual files in isolation and cannot reconstruct distribution networks. This proposal introduces ChainMark, a law-enforcement-controlled cryptographic watermarking and payload-mutation architecture designed for use in authorized sting operations. ChainMark embeds a uniquely keyed, steganographic breadcrumb into seeded files that mutates at each forward event, producing a self-documenting chain-of-custody log that maps the topology of a distribution network in real time. Entire networks can be identified and interdicted simultaneously rather than sequentially. The system is designed with a strict key-custody model ensuring only authorized law enforcement personnel can read or generate payloads.

1. Problem Statement

Current federal strategy against CSAM distribution networks faces a structural limitation: arrests are made sequentially, one node at a time. When a single actor is identified and apprehended, the remainder of the network typically disperses, migrates platforms, or rotates keys before investigators can act. The result is a whack-a-mole enforcement cycle that imposes real operational cost without achieving network-level disruption.

PhotoDNA and NCMEC hash-matching are valuable for detecting known content at scale, but they answer the question 'is this file known illegal content?' — not 'who is in possession of it, who gave it to them, and who have they given it to?' The investigative leap from file detection to network topology currently requires extensive manual subpoena chains, cooperative platform disclosure, and

sequential legal process — each step giving the network time to adapt.

ChainMark addresses this gap. Rather than waiting for content to appear in hash databases, it makes the content itself the investigative instrument.

2. Prior Art and Differentiation

System	Function	Network Mapping?	Limitations
PhotoDNA (Microsoft/NCMEC)	Perceptual hash matching of known files	No	Passive; identifies files, not actors or chains
NCMEC CyberTipline	Platform-reported hash matches	No	Requires platform cooperation; no propagation trace
Traditional Watermarking	Static ID embedded in media	Partial (single hop)	Payload does not mutate; cannot trace multi-hop chains
FBI Sting Ops (existing)	Undercover distribution	Partial	Manual, resource-intensive; not scalable to full network
ChainMark (proposed)	Mutating cryptographic breadcrumb chain	Yes — full topology	Requires seeded origin; cannot passively infect existing networks

3. ChainMark Architecture

3.1 Core Concept

ChainMark operates on a controlled-seeding model. Law enforcement, acting under lawful authority, introduces a specially prepared file into a target network. The file contains a steganographic payload invisible to human observers and resilient to standard re-encoding operations. Unlike static watermarks, the ChainMark payload is designed to mutate at each distribution event — each time the file is forwarded, the receiving client (if running a ChainMark-aware relay, or via a paired device-level mechanism in warranted scenarios) appends a signed event record before passing the file along.

3.2 Payload Structure

Each ChainMark payload consists of three components embedded via spread-spectrum steganography:

Component	Description	Size (approx.)
Origin Token	Signed identifier of the seeding operation, encrypted under FBI master key	256 bits

Event Chain Log	Append-only list of signed forward events; each event: timestamp + node identifier hash	Variable (grows per hop)
Integrity MAC	HMAC over full payload; detects tampering; recalculated at each mutation event	256 bits

3.3 Mutation Protocol

The mutation mechanism is the critical innovation. On each distribution event, the following sequence executes at the relay layer (whether that is a warranted device agent, a cooperating platform integration, or an undercover-operated relay node):

Step 1 — Extract: The current payload is extracted from the steganographic channel using the ChainMark read key. Step 2 — Append: A new event record is constructed: {timestamp, node_id_hash, predecessor_MAC}. Step 3 — Re-sign: The updated event chain is signed with the relay node's ephemeral key (issued by FBI PKI). Step 4 — Re-embed: The mutated payload is embedded back into the file via spread-spectrum steganography. Step 5 — Forward: The modified file is passed to the next node. The relay node logs the event to the central ChainMark registry.

CHAIN RECONSTRUCTION: At any point, law enforcement can extract the full event chain from any recovered file, producing a complete, cryptographically signed map of every node the file passed through — in order, with timestamps. This enables simultaneous warrants across all identified nodes.

3.4 Steganographic Robustness

The payload must survive common file transformations that occur during distribution: JPEG re-compression, color space conversion, minor cropping, and scaling. ChainMark employs a spread-spectrum embedding strategy in the DCT coefficient domain for JPEG/video content, distributing payload bits across the frequency spectrum in a pattern keyed to the master secret. This approach is robust to compression up to approximately 60% quality reduction and survives 5-10% spatial cropping. For higher-destruction scenarios, a redundant embedding in multiple frequency bands provides additional resilience.

4. Key Custody and Access Control

The security of ChainMark against misuse is entirely dependent on strict key custody. The architecture enforces a three-tier key hierarchy:

Tier	Key Type	Held By	Function
Tier 1	Master Read/Write Key	FBI CEAU (sole custody) Hardware Security Module	Generates seeding keys; can read any payload
Tier 2	Operation Keys	Assigned per operation; custody logged in HSM	Authorizes a specific sting operation seed

Tier 3	Node Relay Keys	Issued ephemerally per warranted device	Signs individual mutation events only
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No entity below Tier 1 can read payload content generated by another operation. Node relay keys expire after 72 hours and are non-renewable without Tier 1 authorization. All key generation and usage events are immutably logged. The master key never leaves the HSM.

5. Honest Assessment of Limitations

This proposal is submitted in good faith and the author declines to overstate its capabilities.

Origin Dependency: ChainMark requires law enforcement to be the seeding origin of the tracked file. It cannot passively infect existing networks or retrofit tracking onto files already in circulation. This limits application to proactive sting operations, not reactive detection.

Relay Cooperation Requirement: For the mutation mechanism to function without a device-level agent, a cooperating relay — either an undercover-operated node or a platform integration — must be present at each hop. Fully air-gapped, manual physical transfers break the chain. The static payload (origin token) still survives and provides value even when mutation fails.

Adversarial Stripping: A sufficiently sophisticated actor with knowledge of ChainMark's existence could strip steganographic payloads by aggressive transcoding (e.g., recompressing at very low quality, printing and rescanning). The system's effectiveness depends in part on targets not knowing the mechanism exists — standard operational security for law enforcement tools.

Legal Framework: Seeding operations raise entrapment considerations that require careful legal review. This is an existing challenge for sting operations generally and is not unique to ChainMark, but must be addressed by DOJ counsel before deployment.

Scope Boundary: ChainMark is a network-mapping tool, not a content-generation or content-distribution system. The nature of the seeded content and all operational decisions remain exclusively within law enforcement authority and legal process. This proposal covers the cryptographic and steganographic architecture only.

6. Reference Implementation Sketch

The following pseudocode illustrates the core ChainMark operations. A full reference implementation in Python is available for review by authorized personnel upon request.

6.1 Payload Embedding (Seed Operation)

```
# ChainMark seed – called once at operation origin def
chainmark_seed(image_bytes, operation_id, master_key): origin_token =
sign(operation_id, master_key) # Tier 1 key event_chain = [] # empty at origin
mac = hmac(origin_token + encode(event_chain), master_key) payload =
encode(origin_token, event_chain, mac) return
spread_spectrum_embed(image_bytes, payload, master_key)
```

6.2 Payload Mutation (Relay Operation)

```
# Called at each authorized relay node def chainmark_mutate(image_bytes,
node_id, relay_key, read_key): payload = spread_spectrum_extract(image_bytes,
read_key) origin, chain, mac = decode(payload) assert verify_mac(origin, chain,
mac, read_key) # tamper check event = sign({'ts': now(), 'node': hash(node_id),
'prev': mac}, relay_key) chain.append(event) new_mac = hmac(origin +
encode(chain), relay_key) new_payload = encode(origin, chain, new_mac)
log_to_registry(operation_id=origin, event=event) # central log return
spread_spectrum_embed(image_bytes, new_payload, read_key)
```

6.3 Chain Extraction (Recovery Operation)

```
# Called on any recovered file to reconstruct the network path def
chainmark_recover(image_bytes, master_key): payload =
spread_spectrum_extract(image_bytes, master_key) origin, chain, _ =
decode(payload) operation_id = verify_and_decode(origin, master_key)
network_path = [decode_event(e) for e in chain] return { 'operation':
operation_id, 'path': network_path, # ordered list of nodes with timestamps
'graph': build_network_graph(network_path) }
```

7. Recommended Submission Path

This proposal is most appropriately directed to the following federal entities for review and potential pilot development:

FBI — Child Exploitation and Human Trafficking Unit (CEHTU) — Primary operational consumer; sting operation authority.

HSI — Cyber Crimes Center (C3) — Parallel jurisdiction; cross-border network investigations.

NCMEC — Technology Coalition — Integration with existing hash infrastructure and reporting pipeline.

DARPA / IARPA — R&D; funding pathway for technical maturation of the steganographic and cryptographic components.

DOJ — CEOS (Child Exploitation and Obscenity Section) — Legal framework review for sting operation compliance.

Given the technical maturity of the underlying components, DARPA and IARPA represent credible funding pathways for a formal pilot program. The FBI CEHTU and HSI C3 are the appropriate first points of contact for operational review.

8. Conclusion

The technical foundation for ChainMark exists today. Spread-spectrum steganography, cryptographic signing, and append-only chain architectures are mature disciplines. What ChainMark proposes is their deliberate integration into a law-enforcement-controlled operational tool purpose-built for network-level interdiction of CSAM distribution rings.

The shift from sequential arrest to simultaneous network interdiction represents a qualitative change in enforcement capability. Sequential takedowns are operationally expensive and give networks time to adapt. Simultaneous network-wide action, made possible by a real-time chain-of-custody map, eliminates that window.

The author's role in this effort ends at the paper and any reference implementation made available for government review. Operational authority, legal authorization, and deployment decisions rest entirely with the relevant federal agencies. This proposal exists to put a viable technical option on the table for the people whose job it is to use it.